

ANCOVA

Some (G)LM procedures

<i>Procedure</i>	<i>Dependent variable</i>	<i>Independent variable(s)</i>
Simple regression	1 continuous	1 continuous
One-way ANOVA	1 continuous	1 categorical*
Multi-way ANOVA	1 continuous	2 or more categorical*
Multiple regression	1 continuous	2 or more continuous
ANCOVA (broad sense)	1 continuous	At least 1 categorical* and at least 1 continuous

*either categorical or treated as a categorical variable.

What is common to all GLMs?

Dependent variable = linear function of independent variable(s) + error

Independent variables can be all categorical, all continuous, or a mixture of both

Residuals: independent, normal and homoscedastic

Parameters (and their SE) obtained by least squares

Why do we fit GLMs?

Infer

- Test hypotheses

Describe/Summarize

- Document and quantify trends

- Estimate effect sizes and precision

Predict

- Predict dependent variable

- We want reliable predictions from simple models

ANCOVA terminology

ANCOVA (i.e. strict/traditional sense): a model with **one** fixed categorical and **one** continuous variable, without interaction (i.e. assuming a common slope). The idea is to test the effect of the categorical variable while controlling for the covariate.

ANCOVA (broad sense): a general linear model (GLM) that has both continuous and categorical independent variables, possibly including interactions among these.

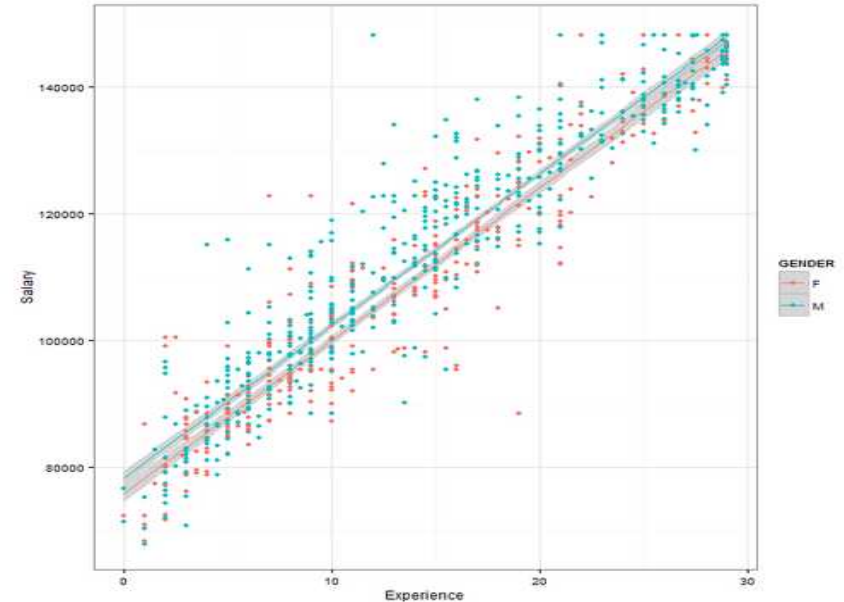
Why use ANCOVA?

Reduce the residual (i.e. error) variation to obtain a more powerful test of the fixed factor:

$$(F_{\text{factor}} = MS_{\text{factor}} / MS_{\text{error}})$$

This allows a test of the differences between factor levels while controlling for the effect of the covariate.

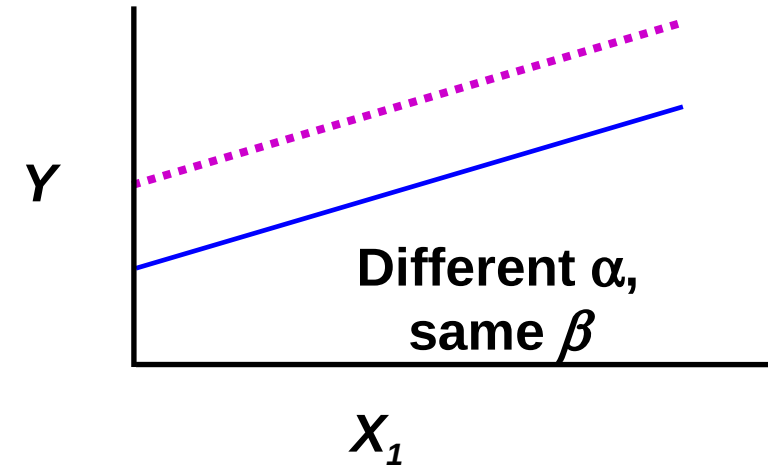
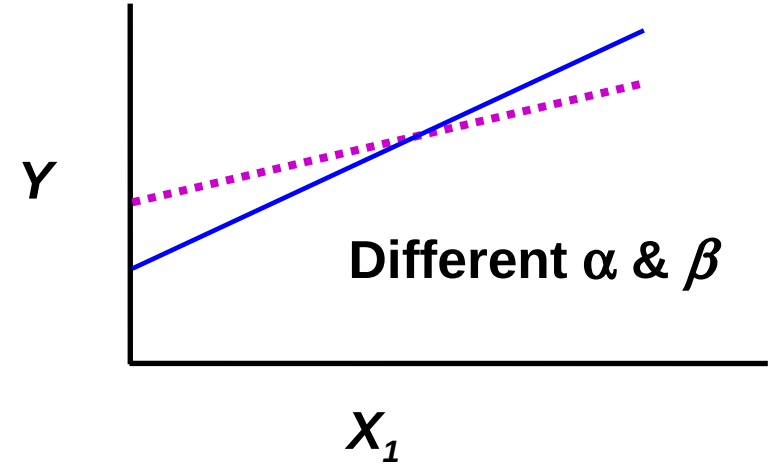
Compare the slopes and/or intercepts of two or more regression lines



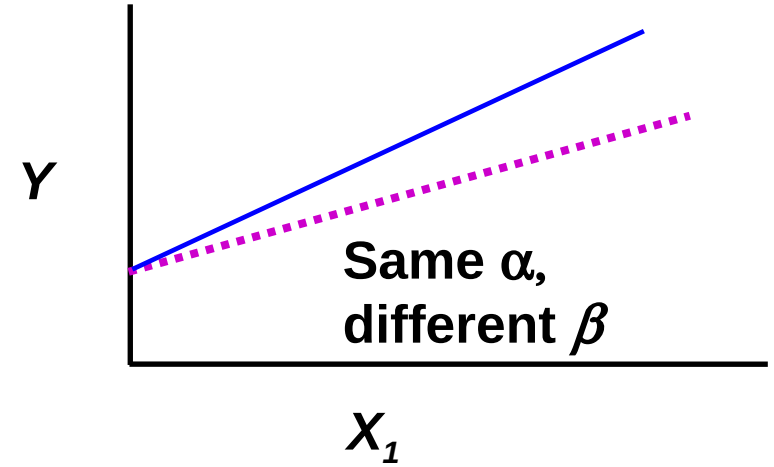
Two linear regressions may have:

differences in both intercepts (α) and slopes (β)

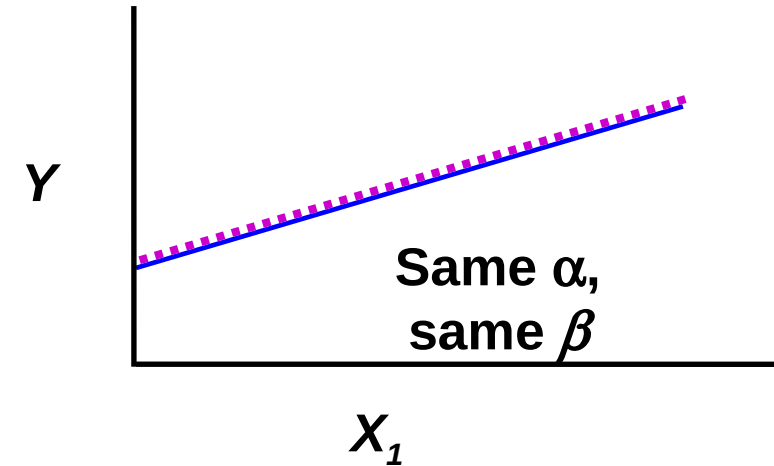
different intercepts but the same slopes



different slopes (β) but the same intercepts (α)



same slopes and intercepts



Null hypotheses

$Y = A + B + A*B$, where A is a fixed, categorical factor and B is a continuous variable

H_{01} (factor A): same as for a single-factor ANOVA

No difference among factor levels in mean Y adjusted for (i.e. controlling for) the covariate (i.e. $\mu_{1(\text{adj})} = \mu_{2(\text{adj})} = \dots = \mu_{k(\text{adj})} = \mu_{(\text{adj})}$)

H_{02} (covariate B): same as for linear regression

No linear relationship between B and Y when controlling for A (i.e. pooled regression coefficient is zero ($\beta = 0$))

H_{03} (interaction)

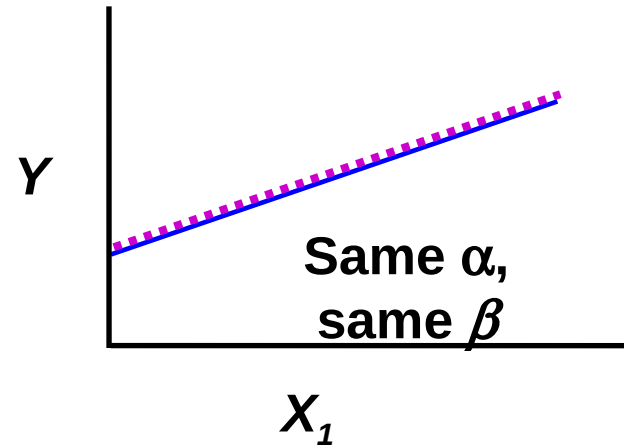
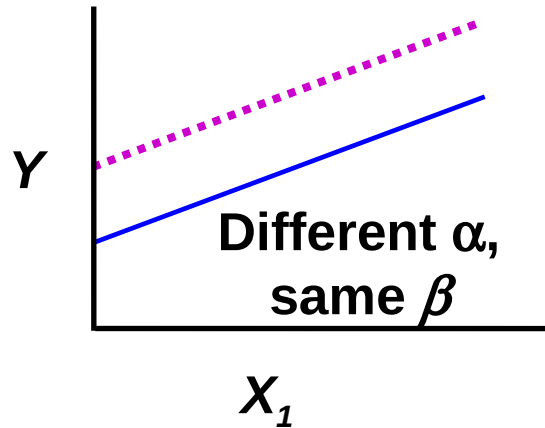
Regression slopes (i.e. of Y on the covariate B) do not differ across different levels of A (i.e. $\beta_1 = \beta_2 = \dots = \beta_k = \beta$)

Traditional ANCOVA

Interested in difference in mean response among levels of the fixed factor and you want to correct/control for the effects of a continuous variable

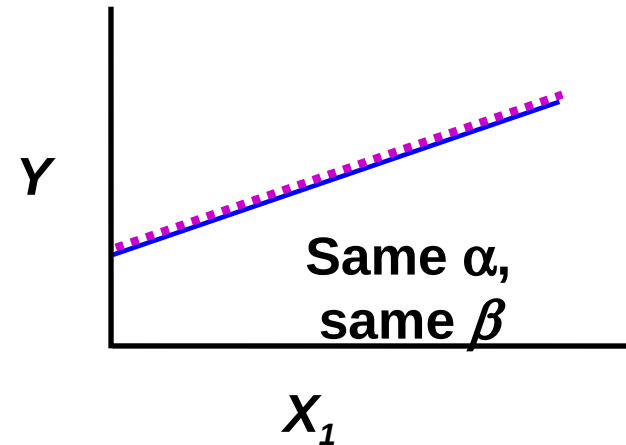
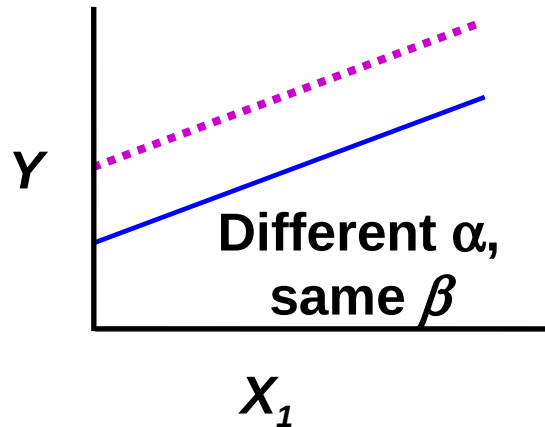
Assumes homogeneity of slopes such that differences in **intercept indicate a difference in mean Y among factor levels**

The first step is therefore to test the interaction, and if non-significant, you drop it and refit the model to test the fixed effect while controlling for the covariate



Traditional ANCOVA

Alternative approach is to just fit the full model (i.e. leave the interaction in) and interpret the type III (i.e. partial) SS tests of each of the parameters, although this has a bit less power.



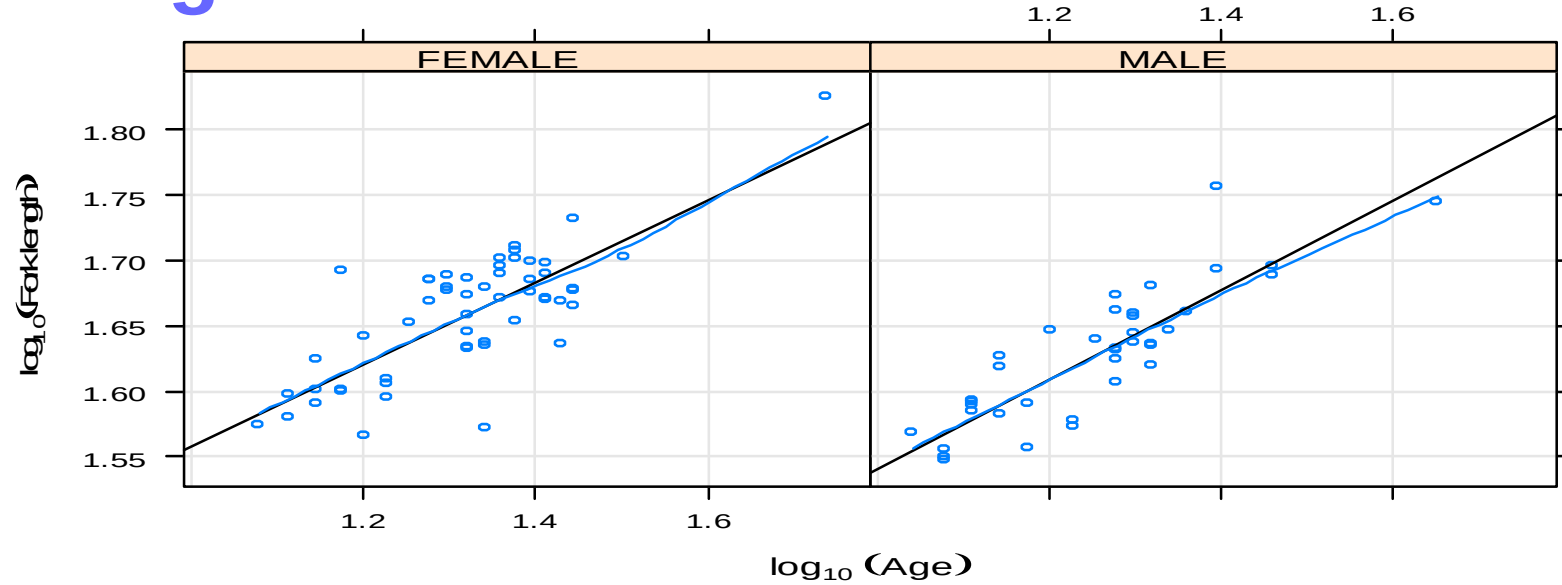
Study on sturgeon

Two goals

Test if growth is the same for males and females

Build predictive model of size as a function of age

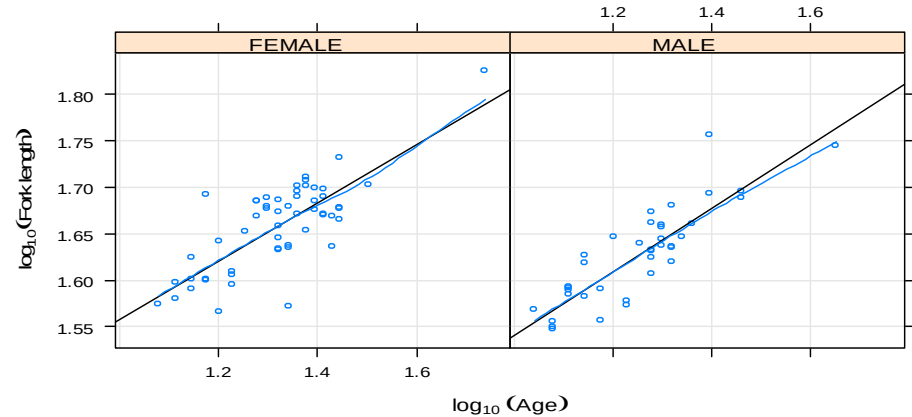
Example 1: effects of sex and age on sturgeon size at The Pas



Analysis

Log(forklength) (i.e. lfkl) is the dependent variable; $\log(\text{age})$ (i.e. lage) is the covariate, and sex is the categorical variable (2 levels).

Q1: is slope of regression of lfkl on lage the same for both sexes (i.e. is there an interaction)?



Effects of sex and age on size of sturgeon at The Pas

```
Anova(model.full, type = 3)
Anova Table (Type III tests)

Response: lfk1
```

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	0.64444	1	794.8182	< 2.2e-16	***
sex	0.00041	1	0.5043	0.4795	
lage	0.07259	1	89.5312	4.588e-15	***
sex:lage	0.00027	1	0.3367	0.5632	
Residuals	0.07135	88			

Conclusion 1: slope of regression of lfk1 on lage is the doesn't differ significantly between the sexes (i.e. no evidence to reject H_{03}) since $\text{Pr}(\text{sex:lage}) > 0.05$

Comparing models with R

```
anova(model.full, model.ancova)
```

Analysis of Variance Table

Model 1: lfk1 ~ sex + lage + sex:lage

Model 2: lfk1 ~ sex + lage

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	88	0.071350				
2	89	0.071623	-1	-0.000273	0.3367	0.5632

Same result (mathematically equivalent)

Q2: is intercept the same for both males and females?

Effects of sex and age on size of sturgeon at The Pas (ANCOVA model, strict sense)

```
Anova(model.ancova, type = 3)
Anova Table (Type III tests)

Response: lfkl

      Sum Sq Df    F value    Pr(>F)
(Intercept) 1.13480  1 1410.1232 <2e-16 ***
sex          0.00149  1    1.8513 0.1771
lage        0.14338  1  178.1627 <2e-16 ***
Residuals    0.07162 89
---
```

Conclusion 2: Mean fork length doesn't differ between males and females when controlling for age (i.e. H_{01} is supported since $\text{Pr}(\text{sex}) > 0.05$)

...and the best predictive model would thus be a simple regression of lfkl on lage, pooling across the sexes.

Assumptions for full model hypothesis testing

Residuals are independent and normally distributed.

Homoscedasticity with reference to both independent variables (i.e. residual variance is equal for all values of covariate X and among levels of the categorical variable).

No measurement error in independent variables

Relationship between Y and covariate is linear.

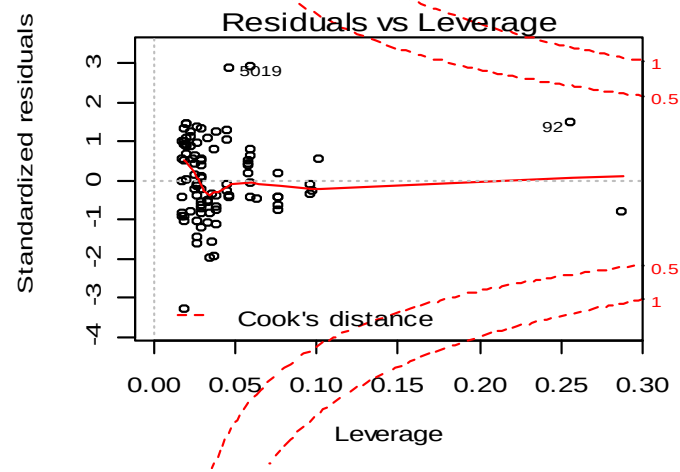
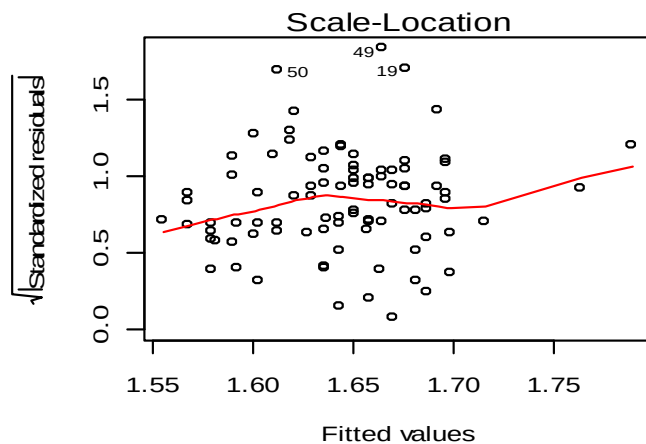
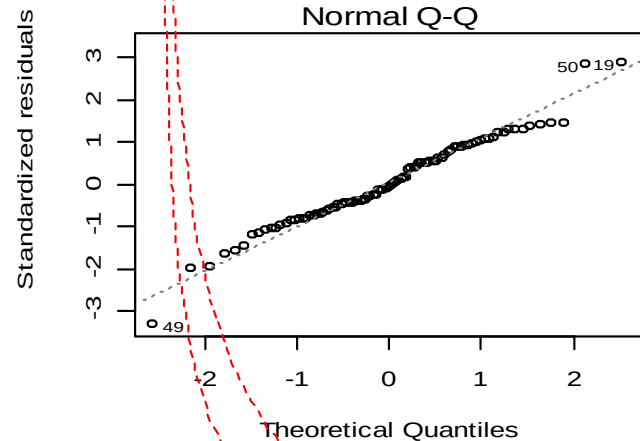
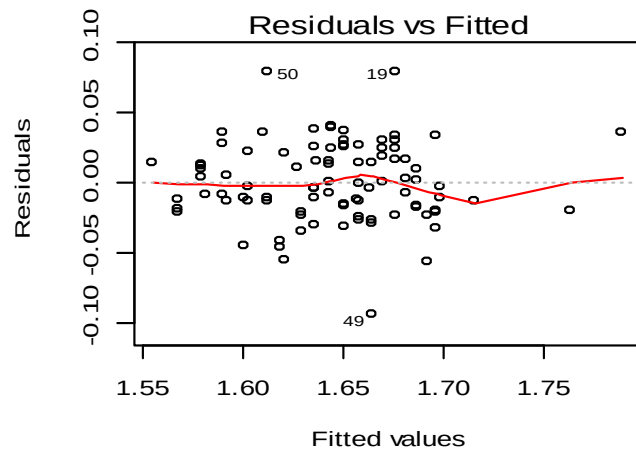
Which residuals do you look at?

Full model?

All models?

Final model?

Full model is what most people do.



Test of Factor A in ANCOVA is a test of intercepts

$Y = A + B + A*B$, where A is a fixed, categorical factor and B is a continuous variable

The test of factor A is often stated as pertaining to differences in the adjusted means

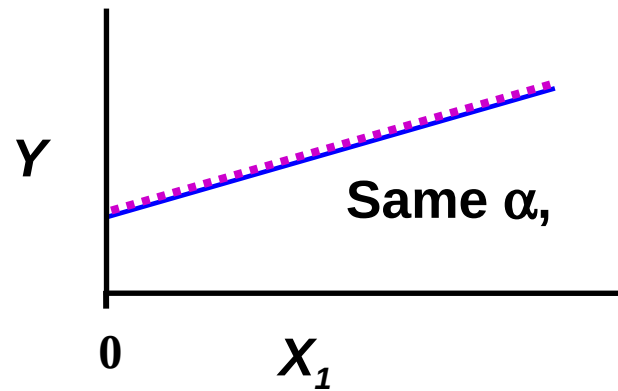
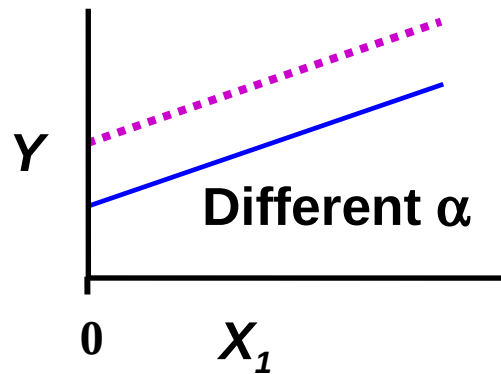
If the regression lines relating Y and B are the same for the different levels of A, this equates to testing whether the Y-intercepts are equal.

The intercept is what is actually tested.

H_{01} (factor A): Y intercepts of the regression lines are all the same

In the presence of an interaction, the main effect of the factor will depend on the scaling of the covariate and extreme caution should therefore be exercised if interpreting it.

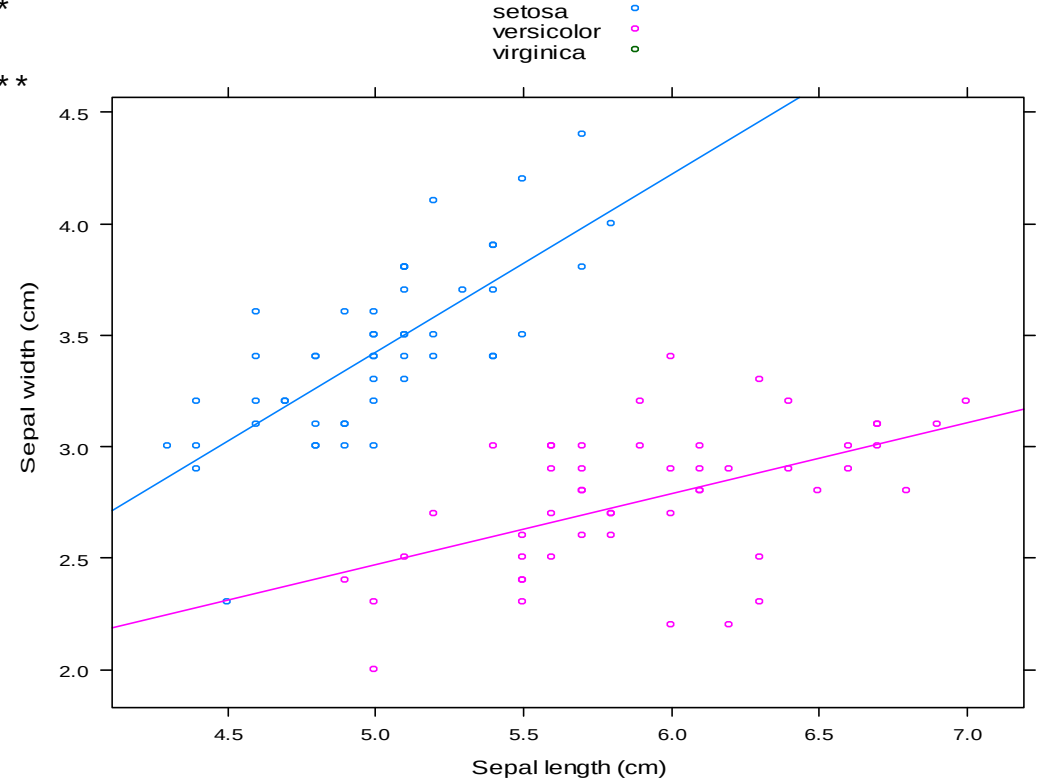
Traditional ANCOVA



```
> Anova(Sepals.lm, type=3)
Anova Table (Type III tests)
```

Response: Sepal.Width

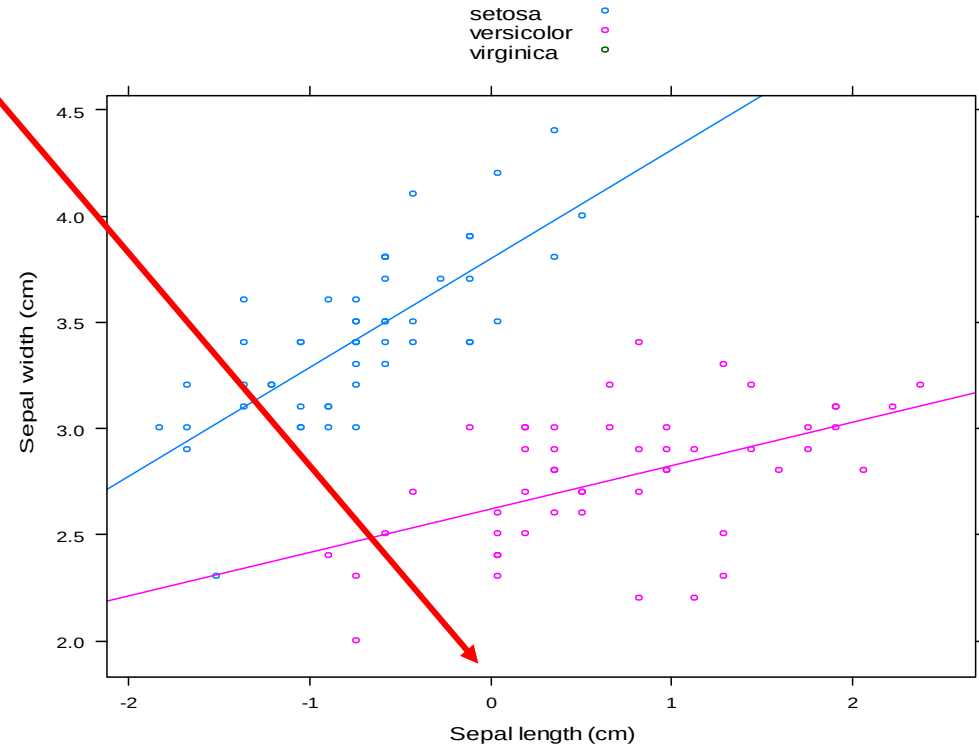
	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.0784	1	1.1319	0.2900487
Sepal.Length	3.8821	1	56.0497	3.41e-11 ***
Species	0.3032	1	4.3769	0.0590690
Sepal.Length:Species	0.9519	1	13.7430	0.0003507 ***
Residuals	6.6492	96		



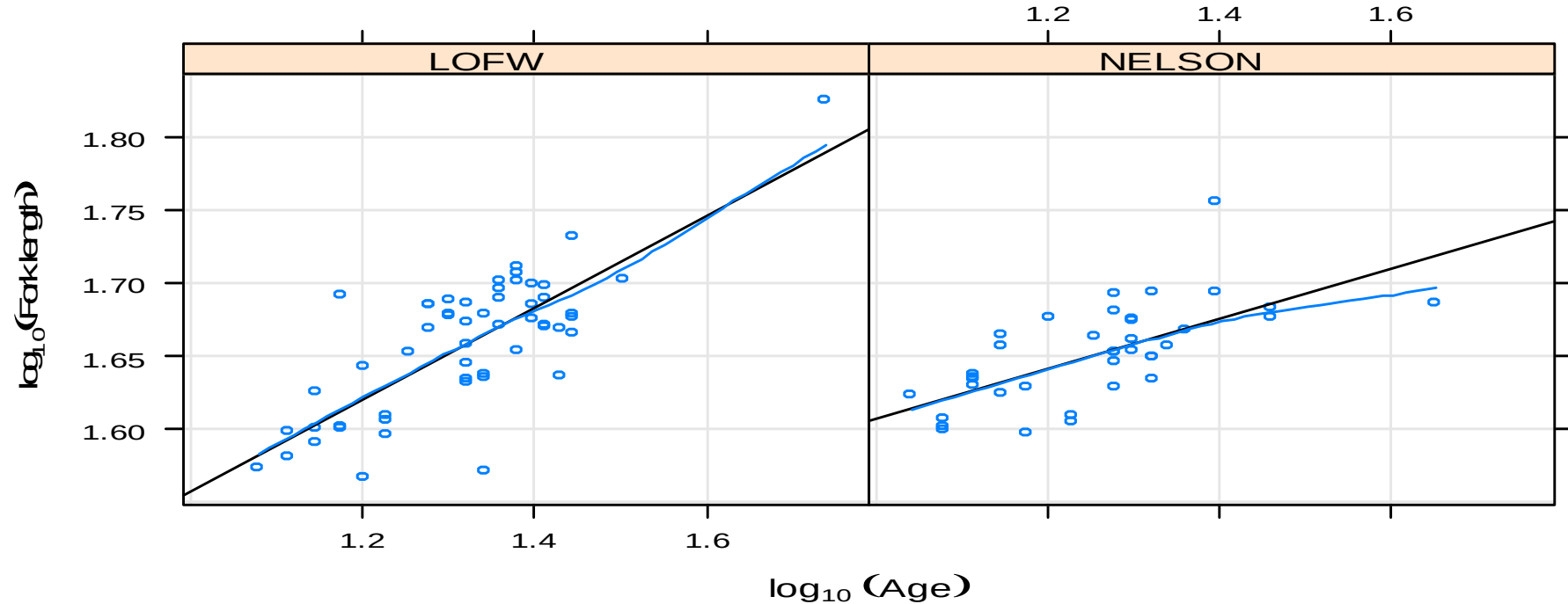
Anova Table (Type III tests)

Response: Sepal.Width

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	260.014	1	3754.059	< 2.2e-16	***
SL	3.882	1	56.050	3.41e-11	***
Species	15.070	1	217.585	< 2.2e-16	***
SL:Species	0.952	1	13.743	0.0003507	***
Residuals	6.649	96			



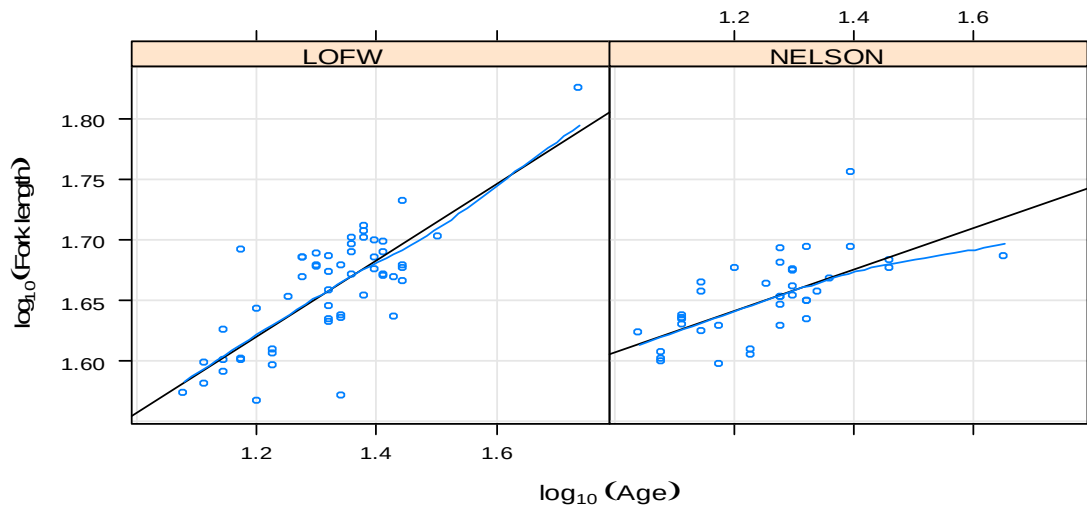
Example 2: Effect of location and age on sturgeon size



Analysis

Log(forklength)(i.e. lfkl) of males is dependent variable; log(age) (i.e. lage) is the covariate, and location (locate) is the categorical variable (2 levels).

Q1: is slope of regression of lfkl on lage the same at both locations?



Effect of location and age on sturgeon size

```
Anova(model.full, type = 3)
Anova Table (Type III tests)
```

```
Response: lfk1
```

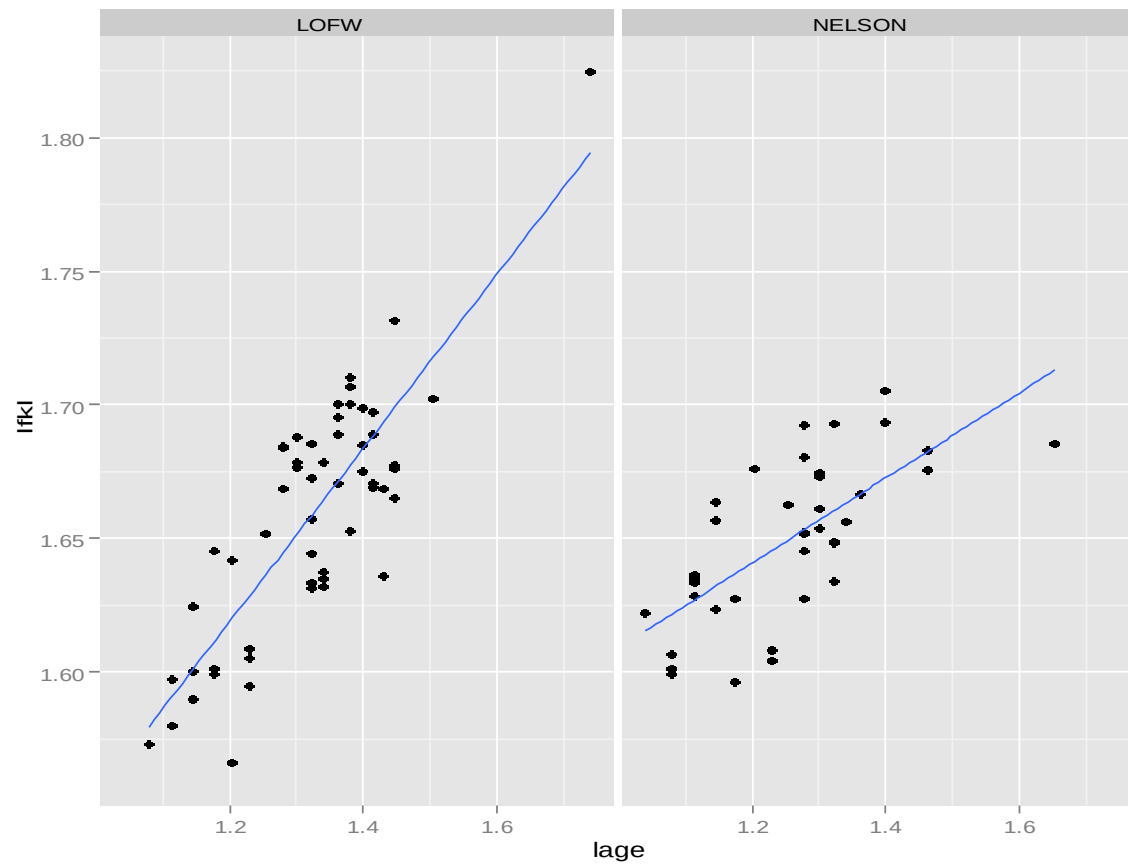
	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	0.62951	1	1078.632	< 2.2e-16	***
lage	0.07773	1	133.185	< 2.2e-16	***
locate	0.00968	1	16.591	0.0001012	***
lage:locate	0.00909	1	15.575	0.0001592	***
Residuals	0.05136	88			

Conclusion: slope of regression of lfk1 on lage is different in the two locations (i.e. reject H_{03}) since $\text{Pr}(\text{lage:locate}) < 0.05$.

So, we can't really control for age when testing for differences in fkl between locations because the relationship between fkl and age differs between locations

should fit separate regressions for each location (this is what the full models is equivalent to).

As with other models, be very careful in interpreting main effects in the presence of an interaction. Plot it!



```
> summary(model.full)
```

Call:

```
lm(formula = lfkl ~ lage + locate + lage:locate, data = mydata)
```

...

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.22840	0.03740	32.843	< 2e-16 ***
lage	0.32529	0.02819	11.541	< 2e-16 ***
locateNELSON	0.22071	0.05419	4.073	0.000101 ***
lage:locateNELSON	-0.16561	0.04196	-3.947	0.000159 ***

$$\text{lfkl} = \mu + \alpha_i \text{lage} + \beta \text{locate} + (\alpha\beta)_i \text{lage:locate}$$
$$\text{Lfkl} = 1.2284 + 0.32529(\text{lage}) + 0.22071(\text{locateNELSON}) - 0.16561(\text{lage})(\text{locateNELSON})$$

For LofW, locateNELSON = 0, so:

$$\text{Lfkl} = 1.2284 + 0.32529(\text{lage}) + 0.22071(0) - 0.16561(\text{lage})(0)$$
$$\text{Lfkl} = 1.2284 + 0.32529(\text{age})$$

For Nelson, locateNELSON = 1, so:

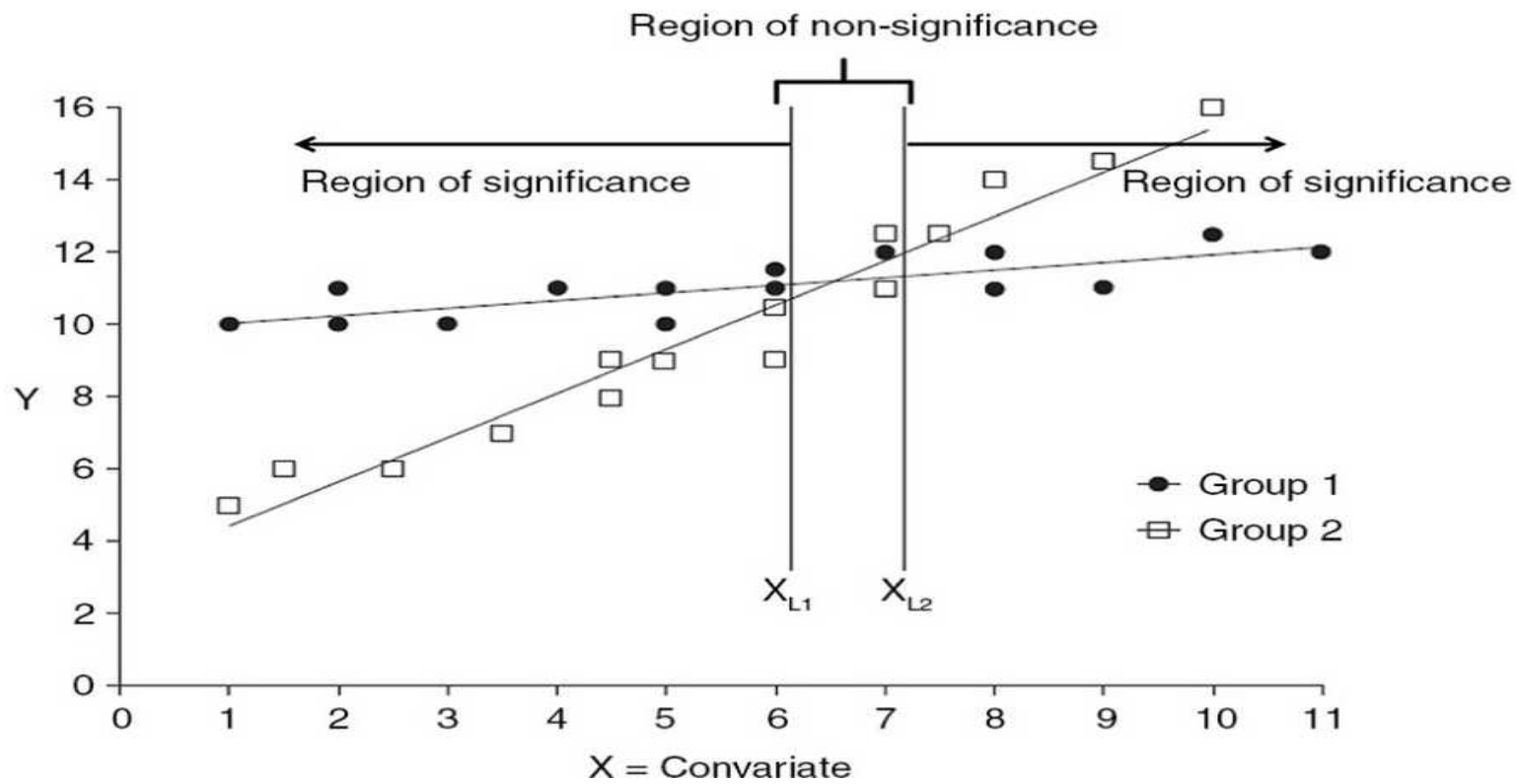
$$\text{Lfkl} = 1.2284 + 0.32529(\text{lage}) + 0.22071(1) - 0.16561(\text{lage})(1)$$
$$\text{Lfkl} = (1.2284 + 0.22017) + (0.32529(\text{lage}) - 0.16561(\text{lage}))$$
$$\text{Lfkl} = 1.4491 + 0.1597(\text{lage})$$

Heterogeneity of slopes

What do you do when slopes differ (i.e. the interaction is significant)?

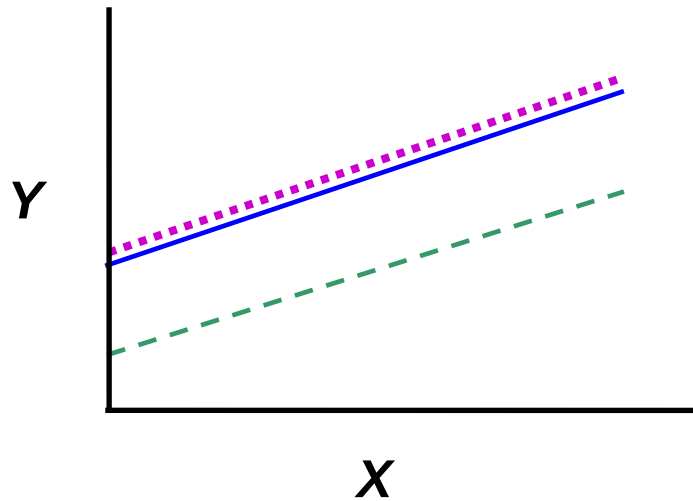
When slopes are heterogeneous, there are a few options that depend on your question of interest:

- 1) If interest is with differences in slopes, then the significant interaction indicates you have this. If there are more than two levels to your categorical variable, then you could do follow-up tests using different pairs of levels to ask which are significantly different (multiple comparison an issue here)
- 2) If the effect of the factor is of interest, there are techniques that allow you to determine the range of the covariate over which the factor levels do vs. do not differ (e.g. Johnson-Neyman procedure)



What do you do if?

More than 2 levels of categorical variable?



Follow same procedure:

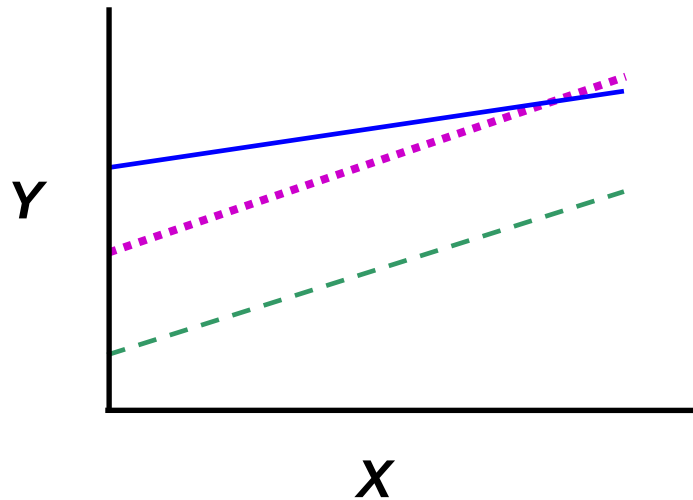
If H_{03} accepted (slopes don't differ), then if H_{01} rejected (i.e. mean Y differs between two or more levels of factor A)

If you're interested in which levels differ than do separate comparisons for all the pairs of levels for A (i.e. subset your data), testing the effect of A in each case.

Always control for family-wise Type I error rate if doing such follow-up tests.

What do you do if?

More than 2 levels of categorical variable?

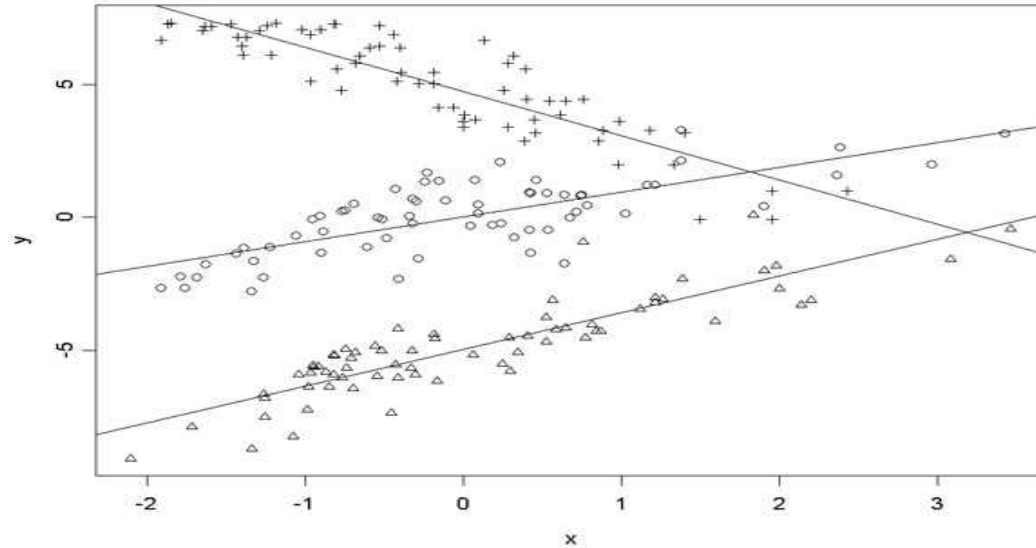


Follow same procedure:

If H_{03} rejected (i.e. slopes differ), then if you're interested in which levels differ in slope then do separate comparisons for all the pairs of levels for A (i.e. subset your data), testing for heterogeneity of slopes in each case.

Always control for family-wise Type I error rate if doing such follow-up tests.

Practice: Which terms are significant? Why?



y : dependent variable

2 independent variables: x
(continuous) and z (categorical)

z has 3 levels (+, o, Δ)

The full model has (and fits
coefficients for) 3 terms:

x

z

$x:z$

A case study

Headline: Male professors make \$8K per year more than females at uOttawa

Male professors make \$8K per year more than females at uOttawa

Call:

```
lm(formula = Salary ~ GENDER, data = anom)
```

Residuals:

Min	1Q	Median	3Q	Max
-47528	-17431	137	15706	40687

Coefficients:

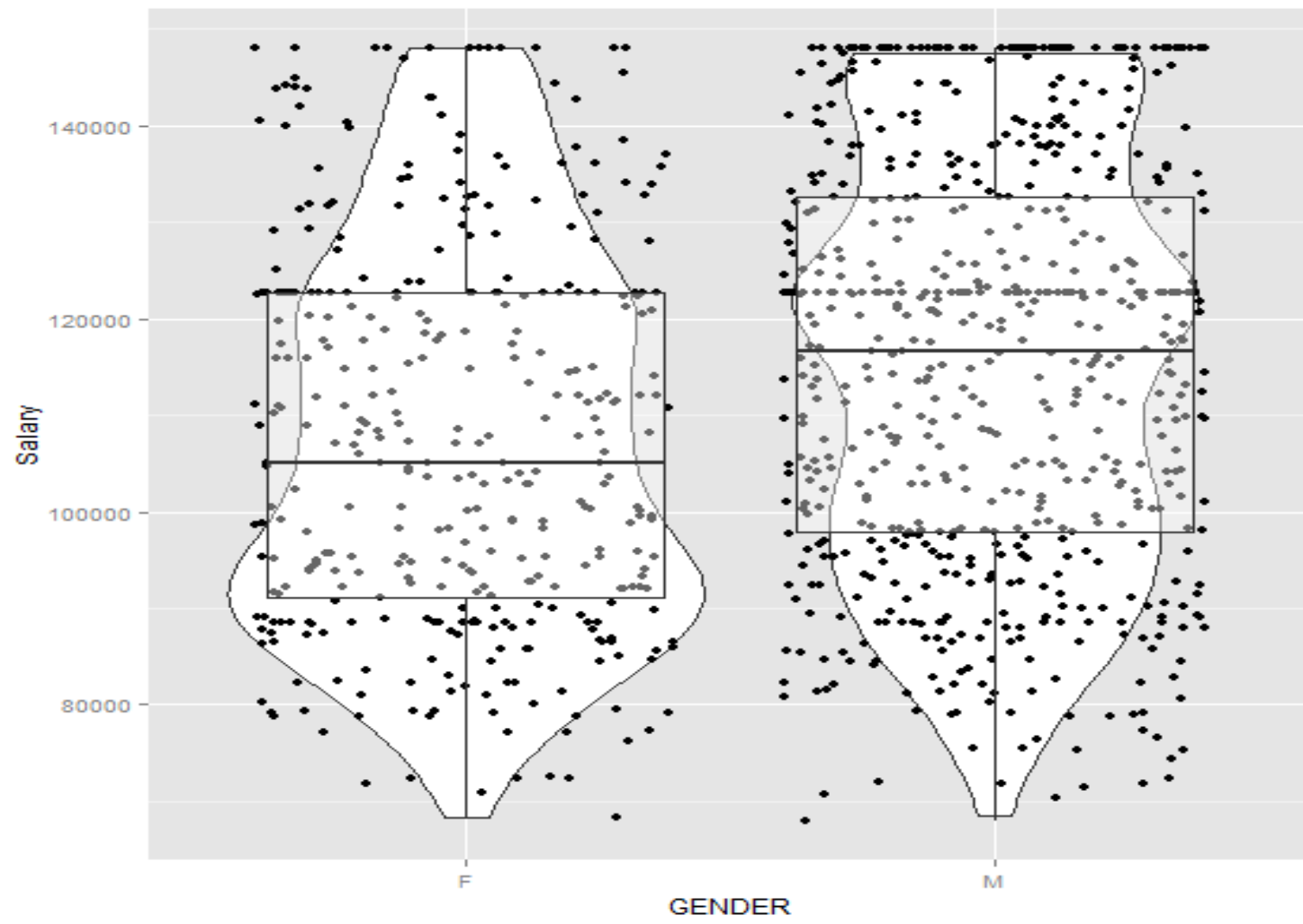
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	107459	1074	100.021	< 2e-16	***
GENDERM	7942	1351	5.877	5.67e-09	***

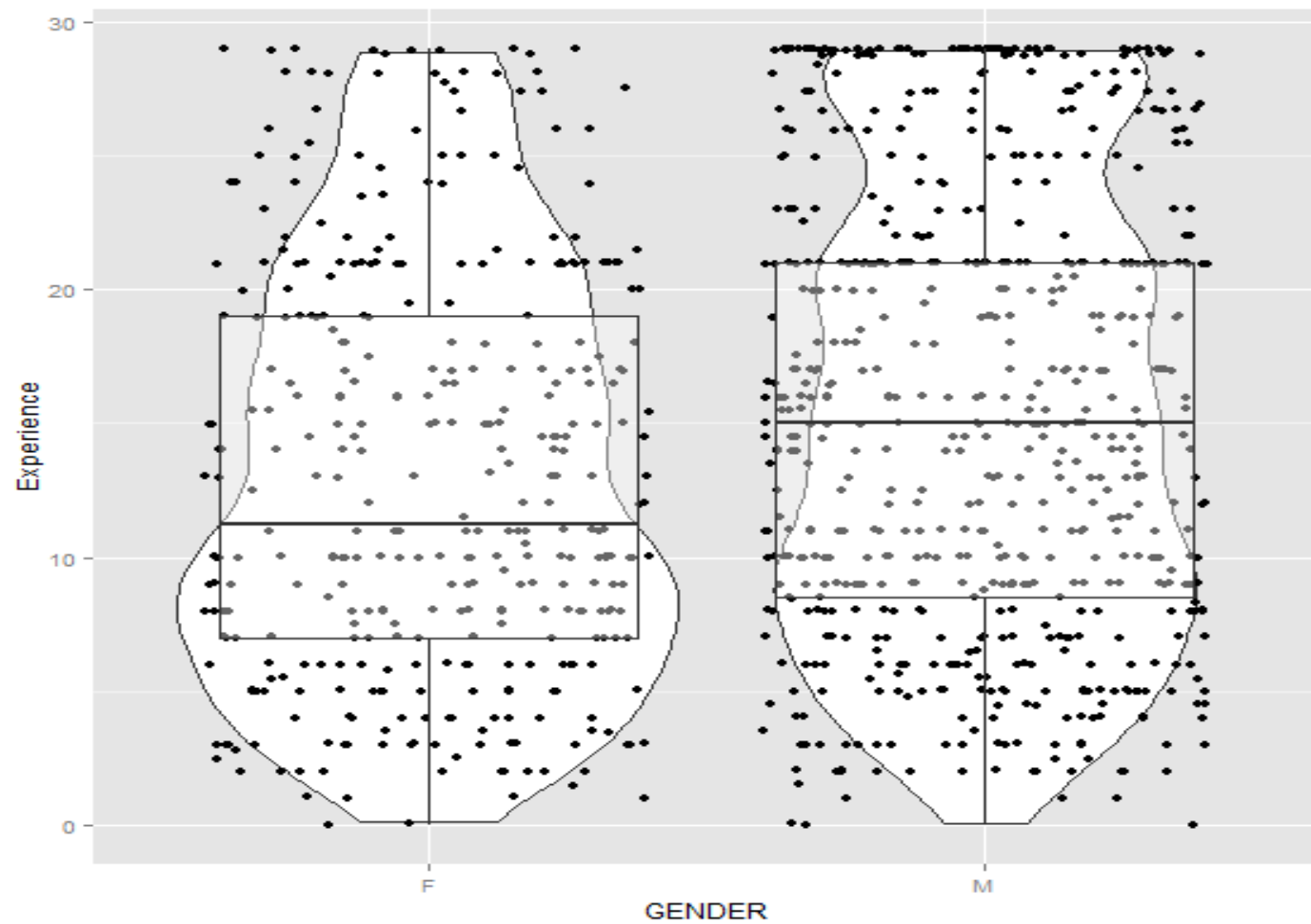
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20720 on 1009 degrees of freedom

Multiple R-squared: 0.0331, Adjusted R-squared: 0.03214

F-statistic: 34.54 on 1 and 1009 DF, p-value: 5.674e-09





Gender bias or experience difference (or both)?

Full Model

Call:

```
lm(formula = Salary ~ GENDER * Experience, data = anom)
```

Residuals:

Min	1Q	Median	3Q	Max
-33202	-4147	-541	3588	40988

Coefficients:

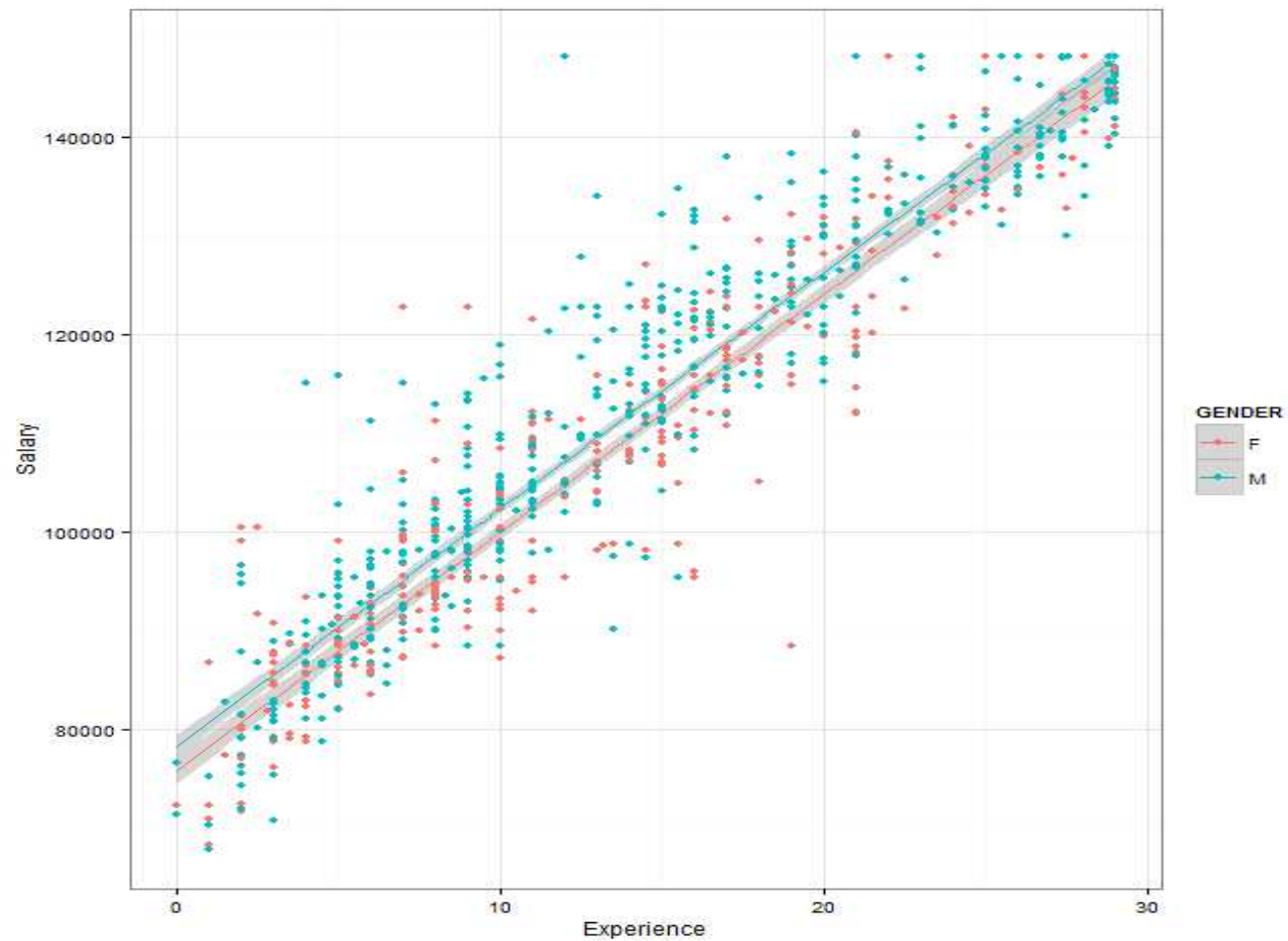
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	75900.54	708.17	107.178	< 2e-16	***
GENDERM	2488.60	911.35	2.731	0.00643	**
Experience	2410.15	46.64	51.672	< 2e-16	***
GENDERM:Experience	-12.74	56.94	-0.224	0.82307	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6913 on 1007 degrees of freedom

Multiple R-squared: 0.8926, Adjusted R-squared: 0.8923

F-statistic: 2789 on 3 and 1007 DF, p-value: < 2.2e-16



ANCOVA (narrow-sense)

Call:

```
lm(formula = Salary ~ GENDER + Experience, data = anom)
```

Residuals:

Min	1Q	Median	3Q	Max
-33152	-4125	-522	3581	41002

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	76012.44	500.97	151.729	< 2e-16	***
GENDERM	2312.02	454.99	5.081	4.46e-07	***
Experience	2401.60	26.74	89.807	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6910 on 1008 degrees of freedom

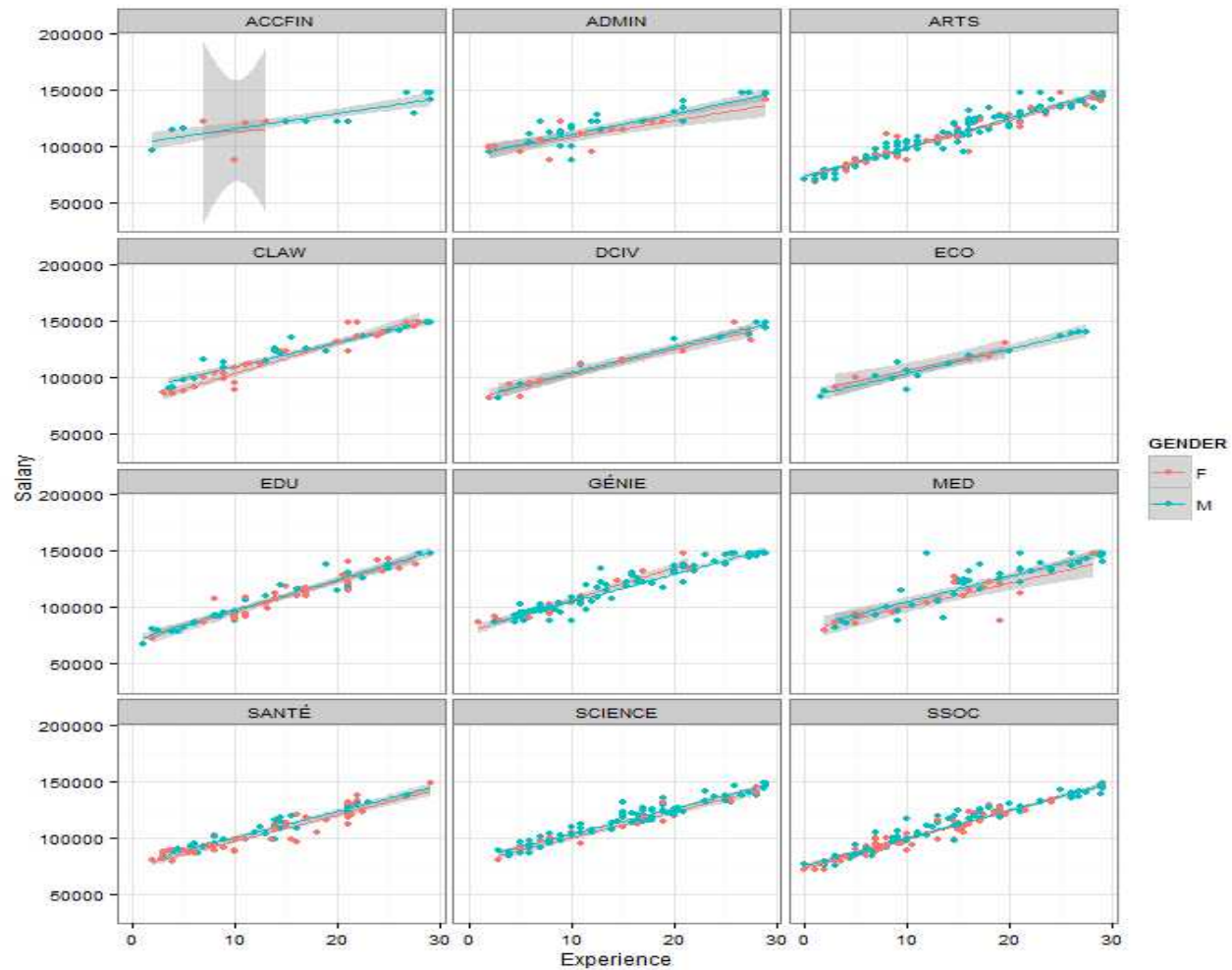
Multiple R-squared: 0.8926, Adjusted R-squared: 0.8924

F-statistic: 4188 on 2 and 1008 DF, p-value: < 2.2e-16

Revised story:

Correcting for experience, male professors make \$2.3K per year more than female professors at uOttawa.

Gender bias or different sex ratios across disciplines (units)?



```
> full2<-lm(Salary ~ GENDER + Experience + Unit, data=anom)
```

```
> Anova(full2, type=3)
```

Anova Table (Type III tests)

Response: Salary

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	1.1984e+11	1	2964.831	< 2.2e-16	***
GENDER	4.8439e+08	1	11.984	0.0005593	***
Experience	3.7084e+11	1	9174.937	< 2.2e-16	***
Unit	7.8349e+09	11	17.622	< 2.2e-16	***
Residuals	4.0298e+10	997			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> summary(full2)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	83952.37	1541.82	54.450	< 2e-16	***
GENDERM	1512.83	437.00	3.462	0.000559	***
Experience	2380.92	24.86	95.786	< 2e-16	***
Unit3ADMIN	-1808.58	1705.08	-1.061	0.289084	
Unit3ARTS	-8957.27	1521.64	-5.887	5.38e-09	***
Unit3CLAW	-2431.46	1689.75	-1.439	0.150481	
Unit3DCIV	-6476.98	1941.14	-3.337	0.000879	***
Unit3ECO	-5796.39	2037.72	-2.845	0.004538	**
Unit3EDU	-10362.06	1669.89	-6.205	8.00e-10	***
Unit3GÉNIE	-3536.42	1587.06	-2.228	0.026084	*
Unit3MED	-5865.13	1647.68	-3.560	0.000389	***
Unit3SANTÉ	-10244.50	1612.07	-6.355	3.17e-10	***
Unit3SCIENCE	-6946.42	1581.75	-4.392	1.25e-05	***
Unit3SSOC	-8852.12	1538.17	-5.755	1.15e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6358 on 997 degrees of freedom

Multiple R-squared: 0.9101, Adjusted R-squared: 0.9089

F-statistic: 776.1 on 13 and 997 DF, p-value: < 2.2e-16

Re-revised story:

**Correcting for experience and discipline,
male professors make ~\$1.5K per year
more than female professors at uOttawa.**

What about rank (i.e. Assistant / Associate / Full professor)? (Unclear what this reflects...)

```
> full2<-lm(Salary~GENDER + Experience + Unit + RANK_CODE, data=anom)
```

```
> Anova(full2, type=3)
```

Anova Table (Type III tests)

Response: Salary

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	1.2466e+11	1	4813.1735	< 2.2e-16	***
GENDER	1.9961e+08	1	7.7068	0.005605	**
Experience	6.2899e+10	1	2428.4893	< 2.2e-16	***
Unit	8.5162e+09	11	29.8910	< 2.2e-16	***
RANK_CODE	1.4553e+10	3	187.2891	< 2.2e-16	***
Residuals	2.5745e+10	994			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	87621.35	1262.97	69.377	< 2e-16	***
GENDERM	973.65	350.72	2.776	0.00560	**
Experience	1755.50	35.62	49.280	< 2e-16	***
Unit3ADMIN	-1364.81	1365.81	-0.999	0.31791	
Unit3ARTS	-11100.32	1221.96	-9.084	< 2e-16	***
Unit3CLAW	-4095.04	1355.27	-3.022	0.00258	**
Unit3DCIV	-7927.69	1558.04	-5.088	4.32e-07	***
Unit3ECO	-7507.10	1633.25	-4.596	4.85e-06	***
Unit3EDU	-11046.29	1338.71	-8.251	4.93e-16	***
Unit3GÉNIE	-6688.92	1278.69	-5.231	2.05e-07	***
Unit3MED	-7684.45	1324.38	-5.802	8.79e-09	***
Unit3SANTÉ	-10293.04	1292.33	-7.965	4.51e-15	***
Unit3SCIENCE	-9190.93	1271.43	-7.229	9.70e-13	***
Unit3SSOC	-10861.73	1235.87	-8.789	< 2e-16	***
RANK_CODEASSISpre2002	-8104.78	1094.72	-7.404	2.82e-13	***
RANK_CODEASSOC	6347.22	496.77	12.777	< 2e-16	***
RANK_CODEFULL	15378.03	753.74	20.402	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5089 on 994 degrees of freedom

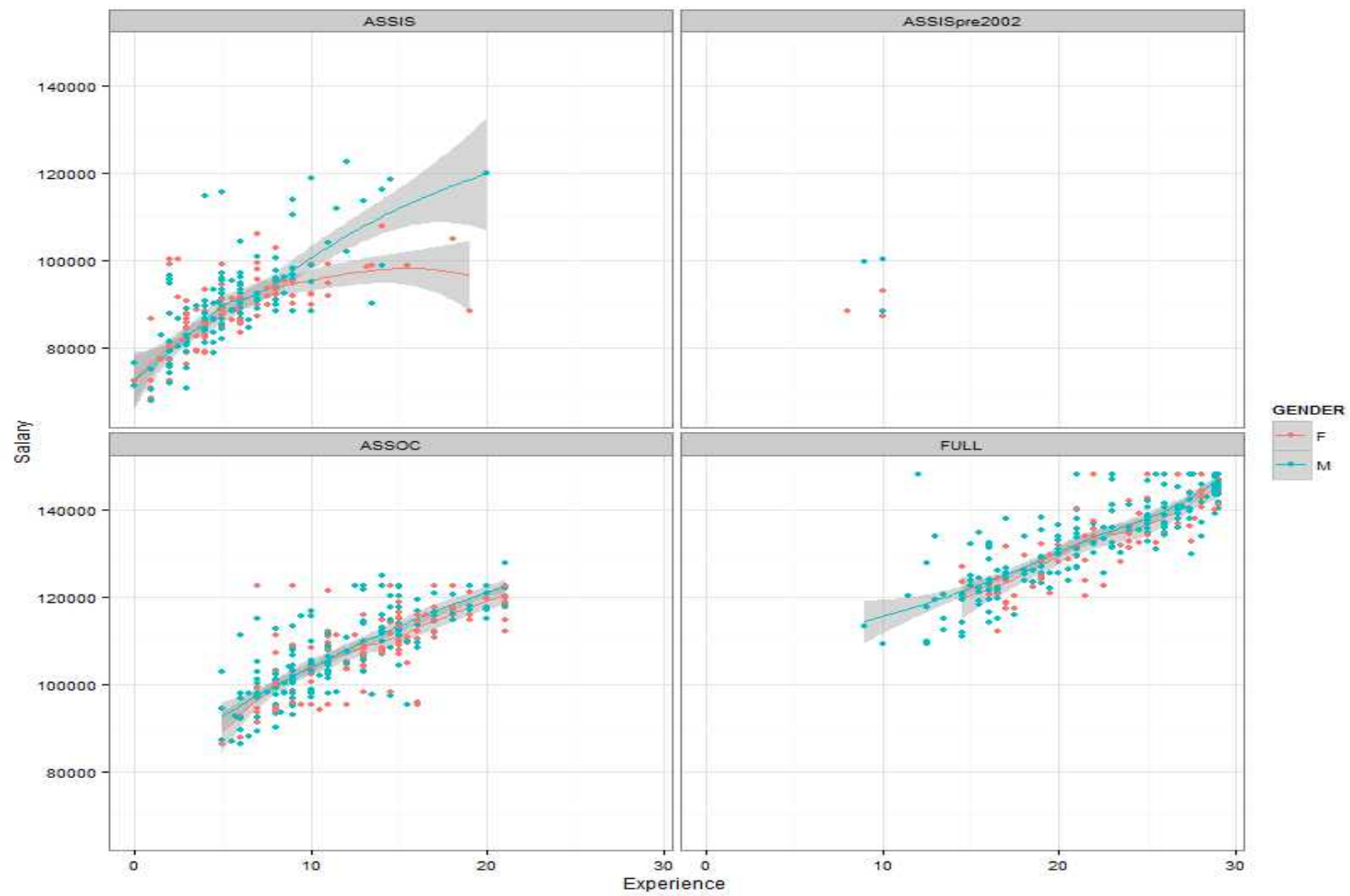
Multiple R-squared: 0.9425, Adjusted R-squared: 0.9416

F-statistic: 1019 on 16 and 994 DF, p-value: < 2.2e-16

Re-re-revised story:

Correcting for experience, rank and discipline, male professors make \$1K per year more than female professors at uOttawa.

What about interactions, non-linearity?



‘Final’ story?

Women salaries are lower than men, on average, by about \$8k (true)

But once discipline and experience are taken into account, the difference shrinks substantially (~\$1.5k), although it remains non-zero.

Taking rank into account further reduces the difference to ~\$970.

Non-linearity/interactions have not been explored and could alter this conclusion.